

Assessing Decision Reliability in Discrete-Event Simulation: A Quantum-Network Case Study

David Vesterlund

Vesterlund Ventures, WestQuant Open

david@vesterlundventures.se

ORCID: 0009-0000-6455-1141

8 October 2026

Abstract

Simulation-based design decisions depend on stochastic variation, analyst choices, and model semantics. We present a workflow that separates these sources of uncertainty using a canonical experiment specification, independent seed pools, explicit intervention gains, and distinct measures of numerical, ranking, decision, and claim agreement. A quantum-network case study compares an author-developed simulator with a harmonized reference model; two framework adapters share the same reference logic and do not constitute independent model replications. Across 400 configurations, a reported two-pool experiment increases label agreement from 71.5% with a single seed to 95.8% with 20-seed averaging. The full protocol and simple averaging achieve the same agreement, so the improvement is attributable to replication rather than a uniquely superior labeling rule. Separate analyses show sensitivity to intervention strength and performance definitions. In a finite-grid memory analysis, 50.8% of 356 matched interior cases differ by at least a factor of four between model variants, with a configuration-clustered 95% interval of 37.4–63.9%. This empirical comparison is conditional on its metric, retained configurations, and shared implementation assumptions. The methodological contribution is an auditable separation of within-model repeatability, between-model sensitivity, and metric incompatibility. We provide reporting requirements and a staged verification plan; full native-protocol and hardware validation remain future work.

Keywords: discrete-event simulation; verification and validation; model uncertainty; decision reliability; reproducibility; quantum networks

1 Introduction

Quantum-network design requires deciding which physical resource to improve first: memory capacity, generation rate, retrieval efficiency, coherence time, or routing. A natural approach is *intervention-based bottleneck identification*: improve each resource by a fixed amount, measure the marginal performance gain ΔG_j , and label the bottleneck as the resource with the largest ΔG_j . This approach is intuitive, model-agnostic, and produces a single actionable label.

But how robust is that label? A reviewer of any such analysis should ask: would the label change with a different random seed? With a larger or smaller intervention? With a different performance metric or fidelity threshold? With a different simulator?

A growing number of discrete-event simulators—NetSquid [1], SeQUeNCe [2], QuISP [3], SimQN [4]—offer different modeling choices for the same physical phenomena. When different models are applied to the same problem, they may produce different numerical predictions. This is not a bug; it is a consequence of the fact that simulation is an abstraction. The question is not whether simulators disagree—they inevitably do—but *which types of scientific conclusions survive a change of simulation model, and which do not*.

This workflow builds on purpose-specific verification and validation of simulation models [5] and explicit computational provenance [6]. Its contribution is to connect those practices to intervention-based decision audits in a shared-code quantum-network case study, while distinguishing repeatability from external validity. Cross-validation of quantum-network simulators provides a domain-specific precedent [7].

1.1 Simulator Output as a Function of Assumptions, Semantics, and Implementation

A simulator’s output depends on three layers:

1. **Model assumptions:** The physical and mathematical models (noise model, generation model, swapping model). These are typically well-documented in publications.
2. **Protocol semantics:** The operational definitions of what constitutes a “completed request,” when a latency clock starts, how waiting and queueing work, when memory slots are locked and released, whether generation is sequential or concurrent, and how deadlines and censoring are handled. These are often *implicit* and may differ between simulators without being documented.
3. **Implementation:** The software engineering (event scheduling, RNG, numerical precision, data structures). These affect reproducibility but not (in principle) scientific conclusions.

This distinction is critical. Two simulators that share the same model assumptions but differ in protocol semantics may produce different scientific conclusions. Two simulators that share both model assumptions and protocol semantics but differ in implementation should target the same output distribution; identical sample paths additionally require coupled random streams and matched event semantics. We therefore write simulator output as:

$$y = f(x, \text{model assumptions, protocol semantics, implementation}), \quad (1)$$

and the trustworthiness of any conclusion depends on which of these factors it is sensitive to. This reframes the question from “which simulator is best?” to “what should a trustworthy quantum-network simulator guarantee for this conclusion to be reliable?”

1.2 Four Types of Agreement

A central conceptual contribution is the distinction between four types of cross-simulator agreement, which are often conflated :

1. **Numerical agreement:** Two simulators produce the same number (or nearly the same number) for a given metric. Measured by MAE, RMSE, or relative error.
2. **Ranking agreement:** Two simulators produce the same ordering of configurations. Measured by Kendall τ or Spearman ρ .
3. **Decision agreement:** Two simulators lead to the same design decision (e.g., the same memory family is selected, the same topology is preferred). Measured by decision-changing disagreement rates.
4. **Scientific-conclusion agreement:** Two simulators support the same scientific claim (e.g., “memory family A outperforms B,” “throughput increases monotonically with memory,” “scalar collapse fails”). Its assessment depends on the specific claim and the required tolerances.

These four types are complementary, not a hierarchy. Exact numerical equality for all compared outcomes entails identical derived rankings under a fixed tie rule, whereas approximate numerical agreement can coexist with decision disagreement. Claim robustness is specific to the stated claim. These types are not interchangeable. Two simulators can have low numerical agreement but high ranking agreement (all values are shifted by a constant factor, but the ordering is preserved). Two simulators can have high numerical agreement but low decision agreement (small differences near a threshold flip the decision). Throughout this paper, each main result is reported on all four levels where applicable.

1.3 Contributions

The contribution is a decision-audit workflow for discrete-event simulation: (i) specify a decision estimand and intervention set before evaluation; (ii) separate Monte Carlo repeatability from model sensitivity; (iii) diagnose metric-definition mismatches before comparing outcomes; and (iv) preserve an auditable mapping from claims to configurations, seeds, code revisions, and outputs. Standard tools such as bootstrap intervals, agreement coefficients, and independent evaluation seeds are combined for this purpose; we do not claim to introduce a new statistical estimator.

The quantum-network case study demonstrates why these distinctions matter. Its scope is two model families with shared code, not a ranking of independent simulator products. The companion memory-physics article addresses hardware-profile and network-design results; this paper addresses the reliability of the procedure used to make those decisions.

1.4 Decision-audit procedure

1. Freeze the configuration set, outcome, interventions, costs, practical gain threshold ε , tie rule, and seed allocation. Record the protocol and analysis revision before evaluating the independent pool.
2. For each configuration and seed, evaluate a baseline and each intervention. Retain paired gains $\Delta G_j = U_j - U_0$; event-level common random numbers require explicit coupling, not merely equal integer seeds.
3. Average gains within each independent seed pool, then select the largest gain using the frozen tie rule. Report no-action cases separately when no gain exceeds ε .
4. Report the fraction with actionable gains, label agreement, contradictory actionable labels, and evaluation loss. An actionable label is not automatically statistically distinguishable from its competitors. Report top-two paired uncertainty separately.
5. Repeat the frozen analysis across intervention strengths, metrics, and documented model variants. Keep these sensitivity axes distinct from seed replication.
6. Resample configurations as clusters when estimating population summaries. Evaluate selected decisions on independent seeds; report loss and its uncertainty alongside categorical agreement. Record the retained population and all exclusions.

The output is a decision audit, not a universal certificate. Equal labels can have different performance, and different labels can have negligible loss. The empirical comparisons below use several campaigns; their run lengths, seed counts, and denominators must not be pooled as though they were one experiment.

2 Setup: Common Experiment Specification and Validation Ladder

2.1 Canonical Experiment Specification

When two simulators are given “the same” scenario, they may still produce different results because the scenario was translated differently. Different RNG implementations generate different arrival traces. Different topology representations assign different node labels. This creates *false disagreement*—apparent simulator differences that are actually translation artifacts.

A Canonical Experiment Specification (CES) reduces avoidable disagreement by specifying every parameter before translation to any implementation. The CES defines:

- **Network:** Topology graph, node count, link distances.
- **Hardware:** Memory family, T_1 , T_2 , retrieval efficiency, readout fidelity.
- **Memory:** Capacity m , slot model, aging model.
- **Links:** Generation probability p_{gen} , attempt time, loss model.
- **Traffic:** Arrival process, arrival rate λ , source-destination pairs, deadline.
- **Protocol:** Swapping probability, purification, scheduling.
- **Routing:** Path selection algorithm.
- **Noise:** Depolarization, amplitude damping, readout errors.
- **Metrics:** Definitions of throughput, fidelity, latency, resource use.
- **Randomness:** Seeds, RNG algorithm, exogenous generation outside simulators.

The proposed CES is simulator-neutral: it does not use any simulator’s parameter names or data structures. Each simulator provides a translator that maps CES parameters to its own configuration format. Exogenous randomness (topology, arrivals, source-destination pairs) is generated outside the simulators to reduce differences in exogenous inputs. Each scenario uses three independent seed streams: (i) a *graph seed* for topology generation, (ii) a *traffic seed* for request arrivals and source-destination pairs, and (iii) a **physics seed** for link generation outcomes, swap outcomes, and decoherence realizations.

2.2 Validation Ladder B0–B6

We propose a benchmark ladder for quantum-network simulation, as a staged verification plan. Table 1 summarizes the ladder and the validation criterion for each level.

The ladder is a proposed dependency structure, not a statement that every stage has passed in this study. Diagnostic higher-level experiments may run after a lower-level mismatch, but they do not then establish cumulative validation. The hardest level (B6) requires claim-level replication across model families. Hardware-calibrated validation against experimental data is a further level that remains future work; we do not have experimental data for contention-dominated scenarios.

B0 analytical verification gate. A finite-deadline single-link reference must include the same attempt count and retrieval semantics as the implementation. With K independent attempts, no contention, and one time-independent retrieval of success probability η , completion probability is $[1 - (1 - p_{\text{gen}})^K]\eta$. A reference of 1 is only an infinite-attempt, loss-free limit. With age-dependent retrieval, the expectation must also average over completion time; multiple required retrievals introduce their joint success probability. A mismatch against a loss-free reference does not by itself identify an implementation error or validate the loss-inclusive model. B0 should therefore be implemented as matched-model component tests before making a cumulative validation claim.

Table 1: Benchmark ladder B0–B6. Failures must be diagnosed before higher-level results are interpreted as validated. B6 in this study is claim-level replication; hardware-calibrated validation remains future work.

Level	Name	Description	Validation Criterion	Scenarios
B0	Analytical primitives	Single-link generation, loss, decoherence	Match analytical rate formulas ($\epsilon_{\text{abs}} < 0.05$)	90
B1	Elementary entanglement	Two-node entanglement generation, memory families	Match analytical or experimental rates	180
B2	Three-node repeater	Single swap, finite memory	Match analytical swapping formulas	360
B3	Multi-hop chain	N -hop chain, path length scaling	Monotonicity, rate decay	180
B4	Finite memory + contention	Memory limit, queueing, multiple requests	Cross-simulator ranking agreement (τ)	200
B5	Heterogeneous topology + traffic	Multiple topologies, stochastic traffic	Structural conclusion agreement	60
B6	Claim-level replication	Full claim grid (C1–C5) across model families	Conclusion-level reproducibility	2,880

2.3 Simulator Frameworks

WestQuant is the native simulator. It models quantum networks with per-node memory capacity m , Bernoulli link generation, depolarizing noise, entanglement swapping, and concurrent request processing with queueing. When all memory slots at a node are occupied, incoming requests queue until a slot frees, incurring queueing delays that affect latency and memory aging.

SeQUeNCe and SimQN (Harmonized Mode) are used via harmonized adapters: their event engines are instantiated, but the physics logic is implemented in a shared harmonization layer using a common Bernoulli-generation, depolarizing-noise, and sequential-blocking model. When memory is full, requests are rejected immediately rather than queued.

Event-trace audit. The SeQUeNCe and SimQN adapters produce bit-identical event traces across all tested scenarios (verified by SHA-256 hashing). The comparison is therefore effectively between one native model (WestQuant: concurrent generation + queueing) and one harmonized reference model (serial generation + immediate blocking), with $S_{\text{models}} = 2$ rather than 3. Treating SeQUeNCe and SimQN as two independent models would constitute pseudoreplication.

SeQUeNCe component pilot. Additional adapter experiments instantiate SeQUeNCe components, but the supplied record does not establish execution of the native entanglement-generation protocol stack. They are excluded from evidence of independent native-protocol validation (Section 8).

2.4 Performance Metrics

All implementations report three primary metrics, computed over the observation window $[t_{\text{warmup}}, T_{\text{obs}}]$ with warmup excluded:

$$A_{\text{req}} = \frac{N_{\text{completed}}}{N_{\text{arrived}}} \quad (2)$$

$$A_{\text{qual}} = \frac{N_{\text{completed,qualified}}}{N_{\text{arrived}}} \quad (3)$$

$$R_{\text{qual}} = \frac{N_{\text{completed,qualified}}}{T_{\text{obs}} - t_{\text{warmup}}} \quad (4)$$

where N_{arrived} is the number of requests that entered the system during the observation window, $N_{\text{completed}}$ are requests that reached the destination within the deadline $D = 10$ s, and

$N_{\text{qualified}}$ are completed requests whose end-to-end fidelity $F \geq F_{\min} = 0.5$. $T_{\text{obs}} = 200\text{ s}$ and $t_{\text{warmup}} = 20\text{ s}$ are reference settings for the cross-model campaigns, not universal settings for every experiment. Bottleneck campaigns explicitly use no warmup and separate request-count limits. The actual stopping rule and retained denominator must be recorded per run. A_{req} and A_{qual} are fractions in $[0, 1]$; R_{qual} is a rate.

2.5 Intervention-Based Bottleneck Identification

Given a network configuration θ and a performance metric $G(\theta)$, we define the *intervention gain* for resource j as

$$\Delta G_j(\theta, \delta) = G(T_{j,\delta}(\theta)) - G(\theta), \quad (5)$$

where $T_{j,\delta}$ is an explicit intervention operator that improves resource j by a relative amount δ . The *soft bottleneck vector* is

$$p_j = \frac{\max(0, \Delta G_j)}{\sum_k \max(0, \Delta G_k)}, \quad (6)$$

and the *dominant bottleneck* is $\arg \max_j p_j$. The bottleneck entropy $H_B = -\sum_j p_j \log p_j$ measures how diffuse the bottleneck is. If all $\Delta G_j \leq \varepsilon$ ($\varepsilon = 10^{-6}$), the configuration is labeled *degenerate*.

We study seven resources: **capacity** (memory slots), **generation** (success probability), **retrieval** (readout efficiency), **coherence** (decoherence rate), **routing** (policy), **deadline** (time limit), and **topology** (edge density).

2.6 Agreement Measures

We use four agreement measures, corresponding to the four types in Section 1.2:

- **Numerical:** MAE or RMSE against reference.
- **Ranking:** Kendall τ for rankings of the soft vector.
- **Decision:** Cohen’s κ for dominant labels and the fraction of changed decisions, with eligibility and degenerate cases reported explicitly.
- **Scientific conclusion:** classification as structural, model-sensitive, or definition-sensitive.

We also report Jensen–Shannon distance for the full soft vector distributions.

3 Noise: Stochastic Fragility

3.1 Stratified Sample

We generate a stratified sample of 400 configurations covering 6 memory families (trapped ion, neutral atom, diamond electron, diamond nuclear, AFC, superconducting), 3 topologies (chain, star, RGG), 3 network sizes ($N \in \{10, 20, 50\}$), 3 traffic levels, and 3 budget levels. The full factor product has 486 cells, so a 400-configuration subset cannot cover every cell. Coverage and sampling weights must be assessed from the configuration manifest. Each simulation run uses 2000 requests (Poisson arrivals), $T_{\text{obs}} = 200$ time units, no warmup, and depolarizing noise at medium level.

3.2 Four Procedures

We assess stochastic fragility with four procedures, each reported on all four agreement levels:

Procedure 1: Label stability. Using 5 seeds per configuration with common random numbers (same seed for both models), we measure label stability as the fraction of seeds agreeing with the modal dominant label. For A_{qual} , the mean stability was 0.686 (WestQuant) and 0.571

(Harmonized). The label changed for 50.7% of configurations across seeds (at least one seed disagreed); 28.2% showed a majority change (stability < 0.5).

Procedure 2: Confidence intervals and statistical ties. To assess whether the 79% “any change” figure at 100 requests reflects genuine instability or insufficient statistical evidence, we computed bootstrap confidence intervals (1000 resamples, 95% level) for the paired difference $\Delta G_{(1)} - \Delta G_{(2)}$ on a subset of 100 configurations with 20 seeds each. A configuration is *statistically tied* if the CI includes zero. Results: 91% showed any change, 58% were statistically tied, and **63% of “any change” configurations were tied**—the data are insufficient to distinguish the top two resources at this sample size. Only 34% showed non-tied label change (label changes AND CI excludes zero).

Procedure 3: Convergence control. We ran 30 configurations with $n_{\text{requests}} \in \{50, 100, 200, 500, 1000\}$ and $n_{\text{seeds}} \in \{5, 10, 20, 50\}$. At 20 seeds, label stability increased from 0.480 (50 requests) to 0.622 (1000 requests)—a 30% improvement. “Any change” decreased from 66.7% to 53.3%; “majority change” decreased from 36.7% to 20.0%. A separate 2000-request experiment with 5 seeds gives 50.7% any-change and 28.2% majority-change. Because the seed count and configuration sample differ, this comparison does not isolate convergence with request count. Residual variation can still reflect Monte Carlo noise, near ties, or transients.

Procedure 4: Performance loss. For the 42 configurations (out of 100) that were not statistically tied, the mean relative loss from choosing the second-best resource instead of the best was 58.5% of the best gain. 62% of non-tied configurations lose more than 50% of the best gain if the second-best is chosen.

3.3 Agreement Levels for Noise

- **Numerical:** Mean label stability 0.686 (WQ) at 2000 requests; MAE in ΔG_j varies by resource.
- **Ranking:** Kendall τ can compare intervention rankings across seeds. Label stability is a categorical decision measure, not a substitute for full ranking agreement; the latter is not reported here.
- **Decision:** 50.7% of configurations show at least one seed disagreement at 2000 requests; 28.2% show majority change.
- **Scientific conclusion:** The residual 50.7% any-change rate at 2000 requests indicates that the bottleneck label is not a stable property of the network at this sample size. The cause of the residual is not yet established: it could be persistent Monte Carlo noise, near-equivalent interventions, rare events, or transients.

3.4 Caveat

The 50.7% figure counts any disagreement, including cases where two resources have nearly identical intervention gains and their ranking flips due to Monte Carlo noise. The broader short-run experiment reported 79.2% any-change. The 63% tied fraction refers to a separate 100-configuration, 20-seed subset with 91% any-change; the denominators differ. The confidence-interval, convergence, and performance-loss analyses were conducted at 100 requests and have not been rerun at 2000; their numbers should be interpreted with this caveat. Common random numbers are imperfect: using the same seed for baseline and intervention does not guarantee that corresponding events use the same random draws when interventions change the event order.

3.5 Independent Seed-Pool Comparison

The reported A/B campaign uses: 400 configurations subsampled from the full $6 \times 5 \times 2 \times 3 \times 5 = 900$ factorial (6 families $\times$ 5 topologies $\times$ 2 network sizes $\times$ 3 traffic levels $\times$ 5 memory budgets),

two disjoint seed pools of 20 seeds each (A: seeds 0–19, B: seeds 20–39), 200 requests per run, and a pilot on 40 configurations to set ε (pilot data excluded from results). The experiment is described as freezing the analysis hash before pool B; that hash and the exclusion ledger must accompany the released data. Each config $\times$ seed runs 1 baseline + 7 interventions (capacity, generation, retrieval, coherence, routing, deadline, topology) = 8 runs, for a total of 128,000 runs.

Pilot. The pilot on 40 configurations $\times$ 5 seeds set $\varepsilon = 0.100$ (median max_delta across interventions). Pilot data is not used in the reported results.

Bottleneck label. The dominant bottleneck is the resource with the largest positive delta-gain. A label is “decided” if it is non-degenerate (max delta $> \varepsilon$). “Agree” means seed pools A and B assign the same label. “Contradict” means A and B assign different non-degenerate labels.

Four procedures $\times$ four metrics. Table 2 reports the results with bootstrap 95% confidence intervals.

Table 2: A/B experiment with actual bottleneck diagnosis: four procedures $\times$ four metrics, with bootstrap 95% CI. 400 configurations, 2 $\times$ 20 seeds, 200 requests, 8 runs per config per seed (128,000 total runs).

Procedure	Frac Decided	Repl Agree	Contradict	Dec Loss
Single seed	0.988 [0.98, 1.00]	0.715 [0.67, 0.76]	0.217 [0.18, 0.26]	0.033
Mean (same budget)	1.000 [1.00, 1.00]	0.958 [0.94, 0.97]	0.043 [0.03, 0.06]	0.006
Mean (convergence)	1.000 [1.00, 1.00]	0.917 [0.89, 0.94]	0.083 [0.06, 0.11]	0.009
Full protocol	1.000 [1.00, 1.00]	0.958 [0.94, 0.97]	0.043 [0.03, 0.06]	0.006

Interpretation. With a single seed, 98.8% of configurations produce an actionable label, but replicate agreement is only 71.5% with a 21.7% contradiction rate—the protocol is not reproducible with one seed. With 20 seeds and the full protocol, 100% are actionable, agreement rises to 95.8%, and the contradiction rate drops to 4.3% with a decision loss of 0.006. This measures repeatability at the tested seed budget; it does not establish correctness or transfer to another model. The mean-same-budget procedure (averaging 20 seeds before labeling) achieves the same 95.8% agreement, confirming that the improvement comes from seed averaging, not from the labeling procedure itself.

4 Analyst Choices: Intervention Strength and Threshold

4.1 Intervention Strength

We tested four relative improvement levels $\delta \in \{0.05, 0.10, 0.20, 0.50\}$, applied uniformly to all resources. At $\delta = 0.05$, 86 of 400 configurations (21.5%) were degenerate (all $\Delta G_j \approx 0$), meaning the intervention was too small to produce a measurable effect. At $\delta = 0.50$, 70 (17.5%) were degenerate. The dominant label changed for 42.0% of configurations across δ levels.

4.2 Metric and Fidelity Threshold

We compared A_{req} and A_{qual} across four fidelity thresholds $F_{\text{min}} \in \{0.30, 0.50, 0.70, 0.90\}$. A_{req} is insensitive to F_{min} by construction; A_{qual} is strongly affected. At $F_{\text{min}} = 0.90$ with A_{qual} , all 400 configurations are degenerate—no request achieves the required fidelity. The dominant label changed for 35.5% of configurations across metric and threshold combinations.

4.3 Agreement Levels for Analyst Choices

- **Numerical:** ΔG_j values shift substantially across δ levels (factor of ~ 10 between $\delta = 0.05$ and $\delta = 0.50$).

Axis 3: Metric and fidelity threshold sensitivity

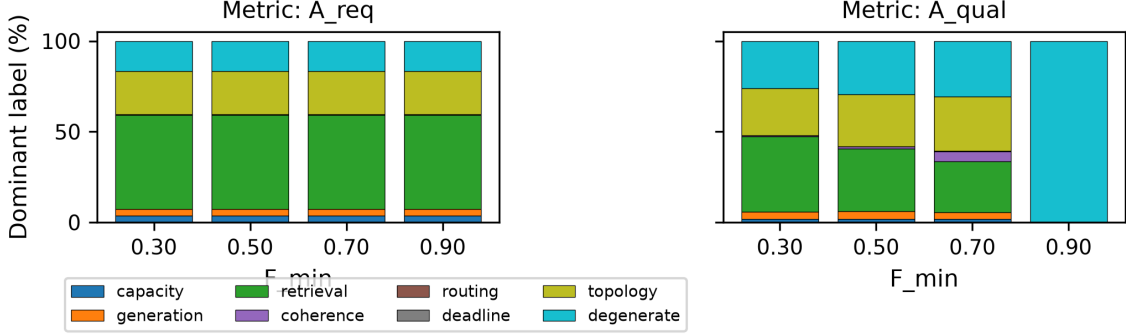


Figure 1: Metric and fidelity threshold sensitivity. Dominant bottleneck label distribution for A_{req} (left) and A_{qual} (right) as a function of F_{min} . A_{req} is insensitive to F_{min} ; A_{qual} becomes degenerate at $F_{\text{min}} = 0.90$.

- **Ranking:** The dominant label changes for 42.0% (intervention) and 35.5% (metric/threshold) of configurations.
- **Decision:** At $\delta = 0.05$, 21.5% of configurations are degenerate—no decision can be made.
- **Scientific conclusion:** The bottleneck label depends on the analyst’s choice of intervention strength and metric/threshold. A label derived at $\delta = 0.50$ with A_{qual} at $F_{\text{min}} = 0.50$ may not transfer to $\delta = 0.10$ or to A_{req} .

4.4 Pairwise Interaction Tests

Saltelli et al. [8] warned that one-at-a-time sensitivity analyses miss interactions between parameters. To address this, we tested all 21 pairwise combinations of the 7 resources on 100 configurations with 5 seeds each (2,100 total measurements). For each pair (j, k) , the interaction effect is $I_{jk} = \Delta G_{jk} - (\Delta G_j + \Delta G_k)$, and the relative interaction is $I_{jk} / \max(|\Delta G_j + \Delta G_k|, \varepsilon)$. Across all 2,100 configuration-pair combinations, 15.8% had relative interaction above the heuristic threshold ($|I_{jk}| / \max(\dots) > 0.1$), and **72% of configurations had at least one resource pair with interaction above the threshold**. The most affected pairs were retrieval + topology (49% of configurations), generation + retrieval (36%), and generation + topology (36%).

5 Model Semantics and Controlled Comparison

5.1 Semantic Dimensions

This study examines differences in *protocol semantics*: how requests are queued, when memory slots are locked and released, whether generation is sequential or concurrent, and how latency is defined. Table 3 summarizes the semantic dimensions for the two systems actually run in this study.

These settings describe the tested adapters and should not be attributed to the native Se-QUENCe or SimQN products. Their contribution to disagreement requires controlled comparison.

5.2 Controlled Semantic-Switch Design

A 2^4 design can vary retrieval-in-delivery, storage-time assignment, blocking policy, and serial versus parallel generation. The response must be defined before decomposition: variance of A_{req}

Table 3: Semantic dimensions for the two systems actually run. “Switchable” indicates whether the dimension can be varied within a single codebase (yes), is a definition mismatch that cannot be compared without harmonization (no), or is partially controllable (partial).

Dimension	WestQuant’s choice	Harmonized adapter choice	Switchable?
Generation mode	Parallel (concurrent link attempts)	Sequential (one link at a time)	Yes
Queueing discipline	FIFO queue (buffer on full)	Reject (drop on full)	Yes
Memory slot locking	Lock on generation attempt	Lock on successful generation	Partial
Slot release	Release on swap completion	Release on swap completion	Yes
Latency definition	Arrival-to-completion (includes queueing)	First-attempt-to-completion (excludes queueing)	No
Deadline and censoring	Count as failures	Count as failures	Yes
Warmup	None	20 s excluded	Yes
Aging model	Continuous decoherence from generation	Continuous decoherence from generation	Yes
Link scheduling	Concurrent across links	Sequential across links	Yes

over variants is different from variance of the difference between two models. If capacity or load also vary, their effects and interactions must be included explicitly.

The draft record describes both a fixed workload ($N = 5$, $m = 2$, $\lambda = 50$) and a broader varying-capacity/load grid for the same variance table. These descriptions do not identify a unique analysis population. Numerical variance fractions and causal closure claims are therefore not used as evidence in this revision. The controlled design remains part of the proposed workflow; its empirical decomposition requires a configuration-level manifest and reproducible analysis before it can support an attribution claim.

5.3 Model-Variant Agreement (Bottleneck Labels)

We compared the WestQuant (concurrent queueing) and Harmonized (sequential blocking) model variants on the same 400 configurations with the same seeds. Of all 400 configurations, the dominant label agreed for 57.5% (230/400). Restricting to the $n = 301$ non-degenerate configurations, the observed agreement is $p_o = 0.532$ and the expected chance agreement is $p_e = 0.407$, giving Cohen’s $\kappa = 0.210$. The mean Jensen–Shannon distance between soft vectors was 0.315.

A separate 60-configuration comparison with 20 seeds reports 51% label agreement despite baseline correlation $r = 0.85$. This illustrates why baseline correlation and intervention-label agreement answer different questions. Results from other retrieval-switch campaigns are not combined with the 400-configuration comparison without a distinct experiment identifier, complete intervention set, and denominator.

5.4 Agreement Levels for Model Choice

- **Numerical:** Mean JS distance 0.315 between WQ and Harmonized soft vectors; MAE in $A_{\text{qual}} = 0.102$ (B6 grid).
- **Ranking:** Kendall $\tau = 0.722$ for A_{qual} (B6), $\tau = 0.430$ for A_{req} ; $\tau = 0.166$ in the B4 contention regime.
- **Decision:** In the 400-configuration comparison, 230 labels agree. Restricting to 301 non-degenerate configurations gives $\kappa = 0.210$; the eligible population differs from the all-configuration agreement denominator.

Table 4: Claim Reproduction Matrix with three-class taxonomy.

Claim	Verdict	Class	Evidence
C1: Memory-family ranking (4 profiles)	REPLICATED	Structural	$\tau = 1.0$, perfect ranking agreement
C2: Topology ranking	REPLICATED	Structural	$\tau = 1.0$, perfect ranking agreement
C3a: Monotonicity of $A(m)$	REPLICATED	Structural	100% monotonicity both models
C3b: Diminishing returns	UNRESOLVED	—	B6 m grid too coarse
C4a: m_c regime-dependence	STRUCTURALLY REPLICATED	Structural	Both agree which families need more memory
C4b: m_c numerical values	MODEL-SENSITIVE	Model-sensitive	$50.8\% \geq 4\times$ disagreement
C5: Scalar collapse failure	STRUCTURALLY REPLICATED	Structural	$R^2 < 0.2$ both models
Latency ranking	NOT COMPARABLE	Definition-sensitive	$\tau = -0.29$, timing definition mismatch
Finite-memory contention	NOT REPLICATED	Model-sensitive	$\tau = 0.17$ in B4

- **Scientific conclusion:** Structural conclusions (rankings, monotonicity) are reproduced; model-sensitive conclusions (m_c , contention) are not; definition-sensitive conclusions (latency) are non-comparable.

6 Where It Matters: Instability, Model Difference, and Boundaries

6.1 Three-Class Taxonomy

We classify observed outcomes into three reporting categories :

6.2 Claim Reproduction Matrix

Table 4 summarizes the three-class taxonomy with the verdict for each claim.

6.3 Claims Reproduced Within the Tested Model Pair

Memory-family ranking. The ordering of four evaluated memory profiles (trapped ion > neutral atom > diamond electron > AFC) by A_{qual} is reproduced across both model families. Kendall $\tau = 1.0$ ($n = 4$).

Topology ranking. The ordering of three topologies (RGG > star > chain) by A_{qual} is reproduced. Kendall $\tau = 1.0$ ($n = 3$).

Monotonicity. Both model families show 100% monotonicity for $A(m)$.

Failure of scalar collapse. All A_{qual} $R^2 < 0.2$ across both model families. The structural conclusion—no scalar provides a strong or universal collapse—is reproduced.

Table 5: m_c disagreement statistics (356 matched interior cases ($m_c < 128$ in both models), $\alpha = 0.95$, A_{qual}). Cluster bootstrap CIs from 2000 resamples at the configuration level.

Statistic	Value	95% CI
Median Δ_m	1.0	—
Mean Δ_m	1.528	—
$\geq 2\times$ ($ \Delta_m \geq 1$)	75.8%	[66.0, 85.1]%
$\geq 4\times$ ($ \Delta_m \geq 2$)	50.8%	[37.4, 63.9]%
$\geq 8\times$ ($ \Delta_m \geq 3$)	33.4%	[21.0, 46.2]%

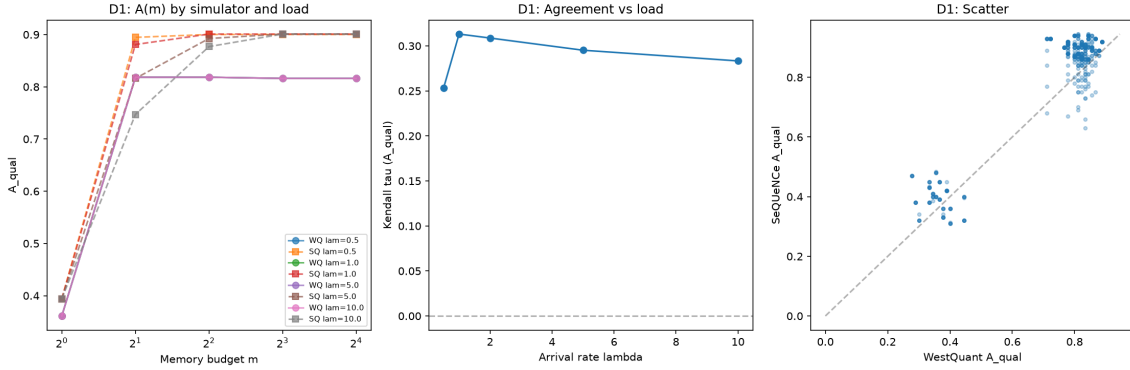


Figure 2: Finite-memory contention disagreement. The reported agreement is low ($\tau = 0.17$). The tested variants differ in queuing and generation semantics; this comparison does not isolate either causal contribution.

6.4 Model-Sensitive Conclusions: Not Reproduced

Relative-grid memory threshold m_c . A sweep over $m \in \{1, 2, 4, 8, 16, 32, 64, 128\}$ contains 3,840 scenario-seed-budget runs per implementation. The analysis code defines $m_c = \min\{m : A_{\text{qual}}(m) \geq 0.95 \max_{m'} A_{\text{qual}}(m')\}$ and excludes curves whose observed maximum is below 0.02. Thus the target is a fraction of the observed maximum, not absolute 95% qualified completion. It also differs from the companion article’s A_{req} threshold. Of 480 nominal curves, 360 paired records enter the comparison; 356 have both thresholds strictly below 128 and four have at least one threshold at 128. The AFC profile is absent from this retained comparison. These are low-signal/retention and grid-boundary issues, not evidence of failure to reach absolute availability 0.95.

Using the log-ratio $\Delta_m = \log_2(m_c^{\text{WQ}}/m_c^{\text{H}})$, the median disagreement is one memory-doubling level, and **50.8% [95% cluster CI: 37.4%–63.9%]** of finite matched cases show $\geq 4\times$ disagreement. The diamond-electron family shows 98.3% $\geq 4\times$ disagreement. Including the four grid-boundary records gives 51.4% [38.1%–64.4%]. Because the threshold is relative to the observed grid maximum, these are finite thresholds, not right-censored observations.

Finite-memory contention. The B4 tier shows $\tau = 0.166$ for A_{qual} —near-zero agreement. This is the regime where model choice most affects conclusions. Feature removal analysis shows that removing the memory limit ($m \rightarrow 128$) gives $\tau = -0.08$ (no restoration), and reducing contention ($\lambda \rightarrow 0.1$) gives $\tau = 0.02$ for $m = 16$ (effectively no agreement).

6.5 Definition-Sensitive Conclusions: Not Comparable

Latency. Latency shows Kendall $\tau = -0.294$ (ranking inversion) across implementations. This comparison uses different timing definitions: WestQuant measures arrival-to-completion (includes queuing time), while the harmonized adapter measures first-attempt-to-completion (excludes queuing). Under high load, queuing time dominates in WestQuant while the adapter

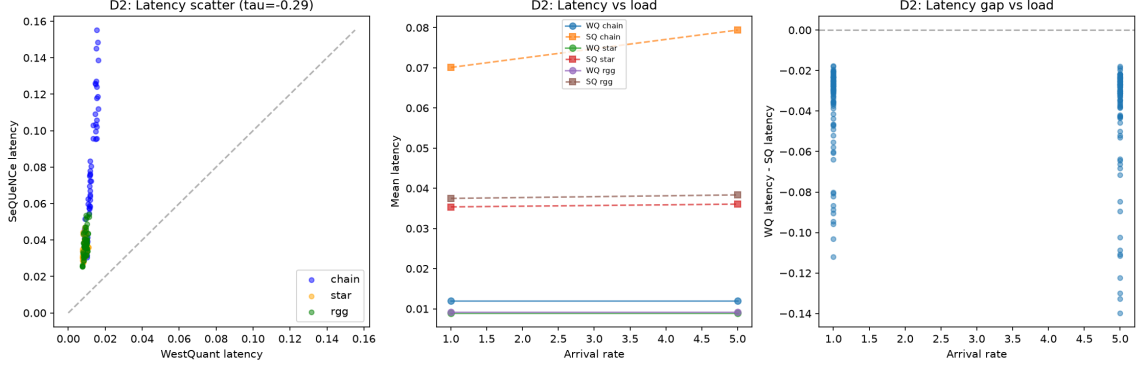


Figure 3: Latency disagreement. The ranking inversion ($\tau = -0.29$) accompanies different timing definitions: WestQuant includes queueing time; the adapter does not.

latency stays flat, inverting the ranking. This is a metric-definition mismatch, not a physical disagreement.

6.6 Regime Dependence and Decision Boundaries

A timescale descriptor such as $\chi_r = \tau_r p_{\text{gen}} / t_{\text{attempt}}$ does not determine model agreement on its own. Offered load, deadlines, queueing semantics, and the distance between competing intervention gains also matter. Small numerical changes near a decision boundary can reverse a label, whereas larger changes may preserve it away from the boundary. The present comparisons do not establish a universal low- χ_r agreement regime or a universal transition point.

6.7 Quality-Qualified vs. Requirement-Qualified Metrics

A_{qual} showed higher ordinal agreement ($\tau = 0.72$) than A_{req} ($\tau = 0.43$) in the B6 grid. This may be because A_{req} is more sensitive to queueing and denominator semantics, while A_{qual} is filtered by a physics-dependent threshold (fidelity) that is more strongly constrained by the harmonized physics. However, the denominator definitions for A_{req} have not been fully harmonized, so this conclusion should be treated as preliminary.

7 Decision Transfer Between Same-Engine Variants

7.1 D6-Capacity: Memory Budget

We test whether capacity decisions made in one model transfer to another. For each context, we select d_A^* in model A using selection seeds, evaluate d_A^* in model B using independent evaluation seeds, and compute regret = $U_B(d_B^*) - U_B(d_A^*)$. The design has 36 contexts, 6 candidate budgets, and 10 selection/evaluation seed pairs, with 258 retained context-seed decisions out of 360 nominal decisions. Candidate evaluations are not separate decisions. The source script skips disconnected selection or evaluation graphs. It uses 100 requests per run, selection seeds 460000–460009 and evaluation seeds 470000–470009; graph realization also changes between selection and evaluation. Crucially, this D6 experiment uses two variants of the same contention simulator: D6-S has serial generation and a 1 s storage cutoff; D6-P has parallel generation and a 10^9 s cutoff. These are not the WestQuant/Harmonized semantics in the preceding adapter comparison. The archived result labels WQ and Harm refer here to D6-S and D6-P, respectively. The reported D6 results are:

- Same choice: 42.2% [32.5, 51.8]

- Regret D6-S→D6-P: 0.047 ± 0.152 [95% CI: 0.007, 0.096]
- Regret D6-P→D6-S: -0.000 ± 0.026 [95% CI: -0.004, 0.004]
- Non-transferable (regret > 0.01): 29.5% [20.1, 39.4] (D6-S→D6-P), 9.3% [5.0, 15.1] (D6-P→D6-S)

The point estimates differ by direction, but a confidence interval for their paired difference is required to establish a statistically significant asymmetry. The D6 variants differ in both generation mode and storage cutoff, so these results cannot isolate the causal effect of generation mode. Estimated loss can be negative because the receiving-model choice is itself selected from noisy data; it is not an exact oracle optimum.

7.2 D6-Placement: Placement Strategy

The placement design uses 12 contexts, 3 strategies, and 10 seed pairs, with 82 retained context–seed decisions out of 120 nominal decisions. Selection seeds are 480000–480009 and evaluation seeds 490000–490009. The same D6-S and D6-P variants are used, with a fixed total memory budget within each context:

- Same choice: 80.5% [61.6, 94.6]
- Regret D6-S→D6-P: -0.001 ± 0.015 [95% CI: -0.003, 0.000]
- Regret D6-P→D6-S: 0.005 ± 0.022 [95% CI: 0.001, 0.010]
- Non-transferable: 4.9% [0, 14.6] (D6-S→D6-P), 12.2% [1.6, 29.3] (D6-P→D6-S)

Placement has higher same-choice agreement than capacity in this particular campaign. This is a descriptive comparison across different candidate sets and contexts, not proof that placement decisions are generally model-independent.

8 Limits of Independent Validation

The harmonized adapters share simulation logic, and the D6 variants use the same contention simulator. Neither comparison establishes independent native-protocol replication. A component adapter can instantiate SeQUeNCe objects without exercising its complete entanglement-generation protocol stack. Different numerical outputs or successful object construction are insufficient evidence of independent model execution.

Pilot and larger component-level comparisons in the draft record use inconsistent run and family denominators. They are not used as quantitative validation here. A future native comparison should archive event traces identifying executed protocol classes, message and quantum-channel events, resource-manager transitions, and the origin of completion counts. It should evaluate the same frozen workloads with harmonized metrics and predeclared tolerances. Hardware validation remains a separate requirement.

9 Standard: A Checklist for Reporting

9.1 Reporting Standard for Bottleneck Diagnoses

We propose that any bottleneck diagnosis should report:

1. **Intervention strength** δ used for each resource.
2. **Number of seeds** and whether common random numbers were used.

Table 6: Trust Stack checklist. Each row identifies a trust level, what it requires, and what this manuscript fulfills.

Level	Name		Requirement	This manuscript
L1	Physics verification		Analytical benchmarks for physical primitives	Matched finite-deadline and retrieval-aware checks are required; a loss-free reference alone is insufficient
L2	Semantic transparency	trans-	Document all protocol-semantic choices	Table 3: 9 dimensions documented for both systems
L3	Canonical loads	work-	Run standardized benchmark ladder	B0–B6 proposed workload design; not a claim that all verification levels passed
L4	Standard traces	event	Export structured event traces	SHA-256 trace hashing for Se-QUeNCe/SimQN identity; format in Appendix
L5	Common experiment specification		Accept CES-format input	CES defined and used across all implementations
L6	Model uncertainty reporting		Report model sensitivity alongside point estimates	τ , κ , Δ_m , cluster bootstrap CIs reported throughout

3. **Performance metric** (A_{req} or A_{qual}) and fidelity threshold F_{min} .
4. **Queueing model** (concurrent, sequential, or other).
5. **Soft bottleneck vector** p_j with uncertainty (e.g., bootstrap CI), not just the argmax label.
6. **Bottleneck entropy** H_B to indicate whether the label is separable or diffuse.
7. **Degenerate cases** reported separately, not assigned an arbitrary label.

9.2 Trust Stack Checklist

We propose a Trust Stack checklist for reporting simulation-based conclusions. Each row identifies what the manuscript itself fulfills (Table 6).

9.3 Recommendations for Reviewers

Based on the findings of this study, we recommend that reviewers of simulation-based quantum-network papers:

1. **Require semantic transparency.** A paper that uses simulation should document the protocol semantics of the simulator, not just the parameter values. If the semantics are not documented, the paper is not reproducible. The semantic-dimensions table (Table 3) provides a template.
2. **Ask whether the conclusion is structural or model-sensitive.** A single model supports a conditional claim. Cross-model robustness must be demonstrated for rankings and monotonicity as well as thresholds and magnitudes. The claim reproduction matrix (Table 4) provides a template for classifying conclusions.
3. **Request cross-simulator comparison for contention-dominated claims.** If the paper claims a capacity-planning result based on a contention-dominated regime, ask whether the result survives a model change. Within the 356 matched interior cases, 50.8% show $\geq 4 \times m_c$ disagreement.

4. **Require provenance.** CES scenarios, event traces, and software versions should be archived as supplementary material. A paper without provenance is a claim without evidence.

9.4 Recommendations for Researchers

1. **Report rankings, not just absolute values.** Selected rankings persisted in this case study, but ranking robustness must be tested for the claim of interest.
2. **Do not compare latency across implementations without harmonizing definitions.** The timing definition must be documented and matched. If the definitions differ, report the difference as a definition mismatch, not a model disagreement.
3. **Treat capacity-planning conclusions as model-dependent.** Report m_c as a range or with model-sensitivity analysis, not as a single number. This study shows that m_c can differ by $\geq 4\times$ in 50.8% of the 356 matched interior cases.
4. **Report the validation level.** State which level of the benchmark ladder (B0–B6) the conclusion claims, and what evidence supports it. A claim at B6 requires cross-model evidence; a claim at B0 does not.
5. **Use cluster bootstrap CIs for cross-simulator comparisons.** Resample configurations (not individual seeds) to avoid pseudoreplication. When CIs include zero, the effect should be reported as not statistically significant.

10 Limitations

1. $S_{\text{models}} = 2$ (**not a true multi-simulator study**). The study effectively compares two model families: the native WestQuant model and one harmonized reference model run through SeQUeNCe and SimQN adapters. SeQUeNCe and SimQN produce bit-identical traces when given the same harmonized model, so they do not constitute independent simulators. Treating them as independent would be pseudoreplication. A full multi-simulator comparison (NetSquid, SeQUeNCe, QuISP, SimQN in native mode) remains open.
2. **Stochastic instability includes Monte Carlo noise.** At 100 requests, the 79.2% “any change” figure overstates the true instability: on a 100-configuration subset, 63% of changes are statistically tied. A separate 2000-request campaign reports 50.7% any-change, but differing seed counts and configuration samples preclude attributing that difference solely to run length. The cause of the residual 50.7% is not yet established.
3. **Common random numbers are imperfect.** Using the same seed for baseline and intervention does not guarantee that corresponding events use the same random draws when interventions change the event order.
4. **No independent native-protocol validation.** The adapters share logic, and D6 compares two parameterizations of one engine. Object instantiation and differing output values are insufficient to establish native-protocol independence.
5. **One-at-a-time interventions.** Single-resource interventions can miss interactions. Pair-wise interaction summaries are exploratory; no validated Sobol variance decomposition is used in this revision.
6. **Semantic attribution remains incomplete.** The available fixed-workload and varying-grid descriptions conflict. A reconciled manifest is required before reporting factorial variance fractions.

7. **Protocol metadata.** The A/B table lacks a complete archived definition of its convergence stopping rule and decision-loss estimator. Its repeatability comparison does not establish a benefit beyond simple seed averaging; algorithm-specific superiority is not claimed.
8. **m_c selection bias.** The main comparison retains 356 paired thresholds below 128 and excludes the AFC profile. Adding four boundary records does not address the missing low-signal population or establish generality across all 480 nominal curves.
9. **No warmup in the bottleneck experiment.** The main bottleneck experiment uses 2000 requests without a warmup period. The convergence study indicates that instability decreases with simulation length; the 2000-request run substantially reduces but does not eliminate the any-change rate.
10. **No hardware validation.** We do not compare against experimental data. Hardware-calibrated validation (hardware-calibrated validation) remains future work.
11. **WestQuant is author-developed.** The WestQuant simulator is developed by the author. It has not undergone independent peer review as a software artifact. The author is both the researcher and the simulator developer, creating a potential conflict of interest. Shared-code comparisons limit independence; a public reproducibility deposit and external validation are needed.

11 Conclusion

Simulation decision audits should separate repeatability under random seeds, sensitivity to analyst choices, model dependence, and metric-definition incompatibility. The case study illustrates that seed averaging can substantially improve label repeatability without establishing correctness in another model. The full protocol matches the same-budget averaging baseline; no additional labeling advantage is demonstrated. Relative-grid memory thresholds vary between the tested model variants, and selected family and topology rankings persist only within the evaluated comparisons.

The contribution is a reproducible audit structure with explicit estimands, campaign identifiers, independent selection/evaluation seeds, and claim-level provenance. Its general components apply beyond quantum networks, but empirical validation here remains confined to this domain and shared-code model variants. Full native-protocol execution and a completed public data release are needed to strengthen the evidence.

Data Availability

The manuscript and a dedicated repository scaffold are prepared for public release. Raw results, the frozen simulator, campaign manifests, dependency lockfiles, and analysis scripts have not yet been deposited in the dedicated public repository. The development repository is private. A completed public release with code revision, checksums, and an archival identifier is required to support the reproducibility claims.

Author Contributions

D.V. conceived the study, designed and implemented all simulators and adapters, ran all experiments, performed all analyses, and wrote the manuscript.

Table 7: Proposed standard event-trace format.

Field	Description
timestamp	Simulation time of event
scenario_id	CES scenario identifier
request_id	Unique request identifier
node_id	Node where event occurs
event_type	generate, swap, deliver, reject, timeout, ...
memory_slot	Memory slot index (if applicable)
link_id	Link identifier (if applicable)
fidelity_before	Fidelity before event
fidelity_after	Fidelity after event
resource_state	Memory occupancy, queue length
cause	Reason for event (if applicable)

Competing Interests

The author is the founder of Vesterlund Ventures and WestQuant Open, and developed the WestQuant simulator used in this study. The compared model variants share the author’s physics code; no truly independent implementation was available at scale. This conflict is a limitation of the study.

Funding

This research received no external funding. Computational resources were provided by the author’s personal hardware.

Declaration of Generative AI and AI-Assisted Technologies

Anthropic Claude, OpenAI ChatGPT, and Cognition Devin assisted with code generation, manuscript organization, and editorial revision. These tools are not authors or independent peer reviewers. The author is responsible for checking the scientific claims, code, data, references, and final text.

A Standard Event Trace Format

A standard event-trace format enables diagnostic comparison: when two simulators disagree, the event traces can be compared event-by-event to identify where and why the divergence occurs. Table 7 shows the proposed format.

In this study, event traces were used to verify the SeQUeNCe-SimQN bit-identity (SHA-256 hashing across all tested scenarios). The trace format is available in the repository and can be used for event-by-event diagnostic comparison when models diverge.

References

- [1] T. Coopmans, R. Knegjens, A. Dahlberg, D. Ferrari, J. Rabbie, and S. Wehner, NetSquid, a network simulator for quantum information using discrete events, *Commun. Phys.* **4**, 164 (2021).

- [2] X. Wu, A. Kolar, J. Chung, D. Jin, T. Zhong, R. Kettimuthu, and M. Suchara, SeQUeNCe: A customizable discrete-event simulator of quantum networks, *Quantum Sci. Technol.* **6**, 045022 (2021).
- [3] R. Satoh, M. Hajdušek, N. Benchasattabuse, S. Suzuki, R. Van Meter, and H. Takesue, QuISP: A quantum internet simulation package, in *2022 IEEE Int. Conf. on Quantum Computing and Engineering (QCE)*, pp. 353–364 (2022).
- [4] L. Chen, K. Xue, J. Li, N. Yu, R. Li, Q. Sun, and J. Lu, SimQN: A network-layer simulator for the quantum network investigation, *IEEE Network* **37**(5), 182 (2023).
- [5] R. G. Sargent, Verification and validation of simulation models, in *Proc. 2005 Winter Simulation Conference*, pp. 130–143 (2005).
- [6] G. K. Sandve, A. Nekrutenko, J. Taylor, and E. Hovig, Ten simple rules for reproducible computational research, *PLoS Comput. Biol.* **9**, e1003285 (2013).
- [7] J. Chung, M. Hajdušek, N. Benchasattabuse, A. Kolar, A. Singal, K. S. Soon, K. Teramoto, A. Zang, R. Kettimuthu, and R. Van Meter, Cross-validating quantum network simulators, in *IEEE INFOCOM 2025 Workshops* (2025), arXiv:2504.01290.
- [8] A. Saltelli, K. Aleksankina, W. Becker, P. Fennell, F. Ferretti, N. Holst, S. Li, Q. Wu, Why so many published sensitivity analyses are false: A systematic review, *Environ. Model. Softw.* **114**, 29–39 (2019).